


```
return self.name
```

2. Apply Migrations

- **Task:** Create and apply migrations to reflect model changes in the database.

1. Generate migration files:

```
python manage.py makemigrations
```

2. Apply the migrations to the database:

```
python manage.py migrate
```

3. Verify the migrations:

- Check the db.sqlite3 file (or your database) for the new tables.
-

3. Register Models in the Admin Interface

- **Task:** Make models manageable through Django's admin interface.

1. Open the admin.py file in the courses app and register the models:

```
from django.contrib import admin
from .models import Course, Lesson, Student
```

```
admin.site.register(Course)
admin.site.register(Lesson)
admin.site.register(Student)
```

2. Start the development server:

```
python manage.py runserver
```

3. Log in to the admin interface at <http://127.0.0.1:8000/admin/> using the superuser account.
 4. Add sample data for Course, Lesson, and Student using the admin panel.
-

4. Query Data in the Django Shell

- **Task:** Use the Django shell to interact with the database.

1. Open the Django shell:

```
python manage.py shell
```

2. Perform the following operations:

- Create a new course:

```
from courses.models import Course, Lesson, Student
```

```
course = Course.objects.create(title = "Python Basics", description
                                = "Learn Python from scratch", duration = 10)
print(course)
```

- Add lessons to the course:

```
lesson1 = Lesson.objects.create(course = course, title
                                 = "Introduction to Python", content = "Python overview...")

lesson2 = Lesson.objects.create(course = course, title
                                 = "Data Types", content = "Explanation of data types...")

print(course.lessons.all())
```

- Add a student and enroll them in the course:

```
student = Student.objects.create(name="Alice Johnson", email="alice@example.com")
student.enrolled_courses.add(course)
print(student.enrolled_courses.all())
```

3. Exit the shell:

```
exit()
```

Lab Deliverables

1. A functional database with tables for Course, Lesson, and Student.
2. Sample data created via the admin interface for:
 - At least 2 courses.
 - At least 4 lessons (2 per course).
 - At least 1 student enrolled in one or more courses.
3. Screenshots of:
 - The Django admin interface showing sample data.
 - The Django shell displaying queries and their results.